# A Robust Converse at Prescribed Degree: Finite Polyhedral Invariants for Fixed-Anchor VBCs

Technical Note

October 12, 2026 — candidate result and proof for review

**Abstract**

At a fixed polynomial degree, a positive-margin safety condition for the convexified monomial transition relation implies a finite polyhedral invariant and a constant-comparison VBC. The condition includes mixing transition pairs and mixing reachable feature vectors at every step. It is equivalent to existence of a VBC with a uniform strict propagation margin, and is sufficient for the usual fixed-anchor VBC class. The proof, its limitation, a same-degree recovery example, and a component-count-independent obstruction are given below.

## 1 Setting and the convexified dynamics

Fix an integer $d \geq 1$. Assume $X \subseteq \mathbb{R}^n$ is nonempty and compact, $f : X \to X$ is continuous, and $X_0, X_u \subseteq X$ are nonempty, compact, and disjoint. All degree bounds below mean *total degree at most* $d$, with $d$ chosen in advance. A forward fixed-anchor VBC consists of finitely many polynomial components $B = (B_1, \dots, B_m)^\top$, a constant entrywise nonnegative $A$, and $\delta_0, \delta_u > 0$ such that

$$B \leq -\delta_0 \mathbf{1} \text{ on } X_0, \qquad B_1 \geq \delta_u \text{ on } X_u, \qquad B(f(x)) \leq AB(x) \quad (x \in X). \tag{1}$$

The same designated component $B_1$ separates *all* unsafe states. No restriction on $\rho(A)$ is assumed. The converse in this note concerns the forward direction.

Let $\phi_d(x) \in \mathbb{R}^D$, $D = \binom{n+d}{d} - 1$, list the nonconstant monomials of degree at most $d$. Constants are represented by affine offsets, so every admissible polynomial is $B_i(x) = \beta_i(\phi_d(x))$ for an affine $\beta_i$. Define

$$\begin{aligned} E &= \operatorname{aff} \phi_d(X), \quad V = E - E, \quad Z = \operatorname{conv} \phi_d(X), \\ I_d &= \operatorname{conv} \phi_d(X_0), \qquad J_d = \operatorname{conv} \phi_d(X_u), \\ \mathcal{F}_d &= \operatorname{conv}\{(\phi_d(x), \phi_d(f(x))) : x \in X\} \subseteq E \times E. \end{aligned} \tag{2}$$

The convex sets $Z$, $I_d$, $J_d$, and $\mathcal{F}_d$ are compact; the affine space $E$ and direction space $V$ need not be bounded. The unit ball $\mathbb{B}_V$ is in $V$, and interiors, distances, and norms are relative to $E$. Omitting the constant feature avoids artificial perturbations of its identically-one coordinate.

A pair $(s, t) \in \mathcal{F}_d$ can be the feature averages of a distribution and its successor distribution, even when $s$ and $t$ are not individual state lifts. An affine comparison on the original graph holds on all of $\mathcal{F}_d$. Thus ordinary lifted trajectory safety does not suffice.

For $\varepsilon \geq 0$, put

$$\begin{aligned} \mathcal{F}_d^\varepsilon &= \mathcal{F}_d + (\varepsilon \mathbb{B}_V \times \varepsilon \mathbb{B}_V), \\ T_\varepsilon(C) &= \{t : \ \exists s \in C, \ (s, t) \in \mathcal{F}_d^\varepsilon\}, \\ C_\varepsilon^0 &= I_d + \varepsilon \mathbb{B}_V, \qquad C_\varepsilon^{k+1} = \operatorname{conv}\big(C_\varepsilon^k \cup T_\varepsilon(C_\varepsilon^k)\big), \\ \mathcal{K}_\varepsilon &= \operatorname{cl} \bigcup_{k \geq 0} C_\varepsilon^k. \end{aligned} \tag{3}$$

The source and successor errors are independent feature-space errors, not a claim about physical state disturbances. There are two convexifications: the transition graph in $\mathcal{F}_d$, and the reachable set at every iteration in (3). Each $C_\varepsilon^k$

is compact and convex; the sets increase and lie in the compact set $Z + \varepsilon\mathbb{B}_V$. Consequently $\mathcal{K}_\varepsilon$ is compact and convex.

Every VBC satisfying (1) forces

$$\mathcal{K}_0 \cap J_d = \varnothing. \tag{4}$$

Indeed, $\{\beta \le 0\}$ contains $I_d$ and is preserved by $\mathcal{F}_d$, convex combinations, and limits, whereas $\beta_1 \ge \delta_u$ on $J_d$. This is a necessary condition; no general converse to (4) is asserted here.

# 2 Candidate converse and its exact scope

**Candidate theorem 1** (Robust convexified safety versus strict propagation)**.** *Under the assumptions above, the following are equivalent.*

1. *There is an $\varepsilon > 0$ such that $\mathcal{K}_\varepsilon \cap J_d = \varnothing$.*
2. *There are finite $m \ge 1$, polynomial $B_i$ of degree at most $d$, a constant $A \ge 0$, and $\delta_0, \delta_u, \eta > 0$ satisfying the separation conditions in (1) and*
$$B(f(x)) \le AB(x) - \eta\mathbf{1} \qquad (x \in X). \tag{5}$$

*Moreover, the robust separation condition in item 1 yields a bounded polyhedron*

$$P = \{z \in E : \beta_j(z) \le 0,\ j = 1, \dots, m\}$$

*with $I_d$ strictly inside it, one designated supporting inequality strictly separating $J_d$, and*

$$(s,t) \in \mathcal{F}_d,\ s \in P \quad \Longrightarrow \quad t + \varepsilon\mathbb{B}_V \subseteq P.$$

*Thus it yields a VBC of the prescribed degree bound in the usual class (1).*

**Scope.** The equivalence is for the additional uniform margin (5). For the usual VBC inequality without $\eta$, the robust separation condition is sufficient but need not be necessary; Section 4 gives a counterexample. The proof is an existence construction over real coefficients. It does not provide a terminating synthesis algorithm or an a priori component bound.

## 2.1 Two finite-dimensional ingredients

**Lemma 1** (Finite outer approximation with a prescribed separator)**.** *Let $K \subseteq E$ be a compact convex body, let $a_1 \in V$ have unit norm, and let $\tau > 0$. There are finitely many unit vectors $a_1, \dots, a_m \in V$ such that, with $h_K(a) = \max_{z \in K} \langle a, z \rangle$,*

$$K \subseteq P = \{z \in E : \langle a_j, z \rangle \le h_K(a_j),\ j = 1, \dots, m\} \subseteq K + \tau\mathbb{B}_V,$$

*and $P$ is bounded.*

*Proof.* Choose an orthonormal basis of $V$ and its positive and negative coordinate directions. Their supporting halfspaces bound a compact box $Q$ containing $K$. Each $y \in Q$ with $\operatorname{dist}(y, K) \ge \tau$ is strictly excluded by a supporting halfspace: if $p$ is its closest point in $K$, take $a = (y - p)/\|y - p\|$, so $\langle a, y \rangle > h_K(a) = \langle a, p \rangle$. The corresponding open exclusion sets cover the compact set $\{y \in Q : \operatorname{dist}(y, K) \ge \tau\}$. A finite subcover, the box inequalities, and the prescribed direction $a_1$ give the result. □

**Lemma 2** (Global row multiplier from a convex relation)**.** *Let $F \subseteq E \times E$ be compact and convex, let $\beta_1, \dots, \beta_m$ be affine, and suppose some $(s_*, t_*) \in F$ has $\beta(s_*) < 0$. If*

$$p_i = \max\{\beta_i(t) : (s,t) \in F,\ \beta(s) \le 0\},$$

*then there is an attained $a_i \in \mathbb{R}^m_{\ge 0}$ such that*

$$\beta_i(t) - a_i^\top \beta(s) \le p_i \quad \text{for every } (s,t) \in F.$$

*Proof.* The closed convex set

$$H_i = \{(\beta(s) + u,\ \beta_i(t) - v) : (s,t) \in F,\ u \geq 0,\ v \geq 0\}$$

contains $(0, p_i)$ on its boundary. It is closed because it is a compact affine image plus a closed cone. A nonzero supporting normal $(w, \gamma)$ can be oriented so that

$$w^\top y + \gamma q \leq \gamma p_i \quad ((y,q) \in H_i).$$

Then $w \leq 0$ and $\gamma \geq 0$, since $u$ and $v$ can increase without bound. If $\gamma = 0$, strict feasibility gives $w^\top \beta(s_*) > 0$ for nonzero $w \leq 0$, contradicting the displayed inequality. Hence $\gamma > 0$; dividing by $\gamma$ and taking $a_i = -w/\gamma$ proves the claim. This is the usual affine-constraint convex duality argument. □

## 2.2 Proof attempt for Theorem 1

*Proof. (1)⇒(2): from the closure to finitely many facets.* Write $K = \mathcal{K}_\varepsilon$. It has nonempty relative interior because it contains $I_d + \varepsilon \mathbb{B}_V$. Compact convex separation of $K$ and $J_d$ provides a unit $a_1 \in V$ and

$$\gamma = \min_{z \in J_d} \langle a_1, z \rangle - h_K(a_1) > 0. \tag{6}$$

Apply Lemma 1 with $\tau = \varepsilon/2$, and define

$$\beta_j(z) = \langle a_j, z \rangle - h_K(a_j), \qquad B_j(x) = \beta_j(\phi_d(x)).$$

Each polynomial has degree at most $d$. Since $I_d + \varepsilon \mathbb{B}_V \subseteq K$ and every $a_j$ is a unit vector,

$$\beta_j(s) \leq -\varepsilon \quad (s \in I_d), \qquad \beta_1(z) \geq \gamma \quad (z \in J_d).$$

The key transition fact is

$$(s,t) \in \mathcal{F}_d,\ \operatorname{dist}(s,K) \leq \varepsilon/2 \quad \Longrightarrow \quad t + \varepsilon \mathbb{B}_V \subseteq K. \tag{7}$$

To prove it, choose $p \in K$ closest to $s$, and $p_\ell \in \bigcup_k C_\varepsilon^k$ tending to $p$. For large $\ell$, $\|p_\ell - s\| < \varepsilon$. For every $w \in \varepsilon \mathbb{B}_V$, the pair $(p_\ell, t + w)$ belongs to $\mathcal{F}_d^\varepsilon$. If $p_\ell \in C_\varepsilon^{k_\ell}$, then $t + w \in C_\varepsilon^{k_\ell + 1} \subseteq K$. This proves (7). In particular, the proof does not assume that passing to the closure automatically preserves every boundary source fiber of the perturbed relation.

For $(s,t) \in \mathcal{F}_d$ with $s \in P$, the outer approximation gives $\operatorname{dist}(s,K) \leq \varepsilon/2$. By (7), $t + \varepsilon \mathbb{B}_V \subseteq K$, hence

$$\beta_i(t) \leq -\varepsilon \quad (i = 1, \ldots, m).$$

There is a strictly feasible source in $\mathcal{F}_d$, obtained from any $x_0 \in X_0$. Lemma 2, applied row by row, therefore supplies $A_i \geq 0$ with

$$\beta_i(t) - A_i \beta(s) \leq -\varepsilon \quad ((s,t) \in \mathcal{F}_d).$$

Stacking the rows and restricting to the original graph yields (5) with $\eta = \delta_0 = \varepsilon$ and $\delta_u = \gamma$. The same argument gives the stated relational invariance of $P$.

*(2)⇒(1): strict propagation tolerates feature errors.* Write the linear part of $\beta_i$ on $V$ as $a_i$, let $L_i = \|a_i\|$, and set

$$L = \max_i L_i > 0, \qquad M = \max_i \left( L_i + \sum_j A_{ij} L_j \right) > 0.$$

The positivity follows from the separating anchor. Choose

$$0 < \varepsilon < \min\{\delta_0/(2L),\ \eta/(2M)\}.$$

Affine linearity extends (5) from the graph to $\mathcal{F}_d$. For $(s,t) \in \mathcal{F}_d$ and $e, w \in \varepsilon \mathbb{B}_V$,

$$\begin{aligned} \beta_i(t + w) &\leq (A\beta(s + e))_i - \eta + \varepsilon \left( L_i + \sum_j A_{ij} L_j \right) \\ &\leq (A\beta(s + e))_i - \eta/2. \end{aligned}$$

Also $\beta \leq -\delta_0 \mathbf{1}/2$ on $C_\varepsilon^0$. Nonnegativity of $A$ and the displayed perturbed comparison imply inductively $\beta \leq 0$ on every $C_\varepsilon^k$; affine inequalities persist under convex combinations and limits. Thus $\beta \leq 0$ on $\mathcal{K}_\varepsilon$, whereas $\beta_1 \geq \delta_u$ on $J_d$. The sets are disjoint. □

*Remark* (Rational approximation). Strict margins on compact $X$ also imply existence of a VBC with rational polynomial coefficients and a rational matrix $A \geq 0$, at the same degree and component count. Indeed, the separation values and the residual $AB - B \circ f$ depend continuously on these finitely many coefficients, uniformly on $X$, since $f(X) \subseteq X$. Sufficiently close rational approximations, keeping $A \geq 0$ and errors below half the respective positive margins, preserve all inequalities. This is an existence guarantee, not a computable rounding procedure for unspecified compact data.

**Corollary 1** (A sufficient condition that can be checked on a chosen body)**.** *It suffices to exhibit a compact convex $C \subseteq E$ and $\varepsilon > 0$ satisfying*

$$I_d + \varepsilon \mathbb{B}_V \subseteq C, \qquad T_\varepsilon(C) \subseteq C, \qquad C \cap J_d = \varnothing.$$

*Then $\mathcal{K}_\varepsilon \subseteq C$, and Theorem 1 constructs a finite degree-$d$ fixed-anchor VBC.*

This is a robust convex-body assumption, not an assumption that a finite polyhedral invariant already exists. The theorem uses supporting halfspaces to obtain finitely many functions and the global comparison matrix.

## 3 Comparison with Horváth–Song–Terlaky, Theorem 3.10

For a prescribed set $S = \{x : g(x) \leq 0\}$, Theorem 3.10 of [1] assumes convex $g$, concave $g \circ f_d$, and a strict feasible point. It characterizes invariance by existence of $\alpha \geq 0$ with

$$g(f_d(x)) \leq \alpha g(x) \qquad (x \in \mathbb{R}^n). \tag{8}$$

Its quadratic case dispenses with the concavity assumption when both functions are quadratic. The proof uses the convex Farkas lemma, or the S-lemma in that quadratic case; Remark 3.11 discusses multiple constraints.

| Issue | HST Theorem 3.10 | Candidate Theorem 1 |
|---|---|---|
| Object | Given convex sublevel set in state space | Constructed finite polyhedron in monomial space |
| Transition model | Deterministic $f_d$ | Convex relation $\mathcal{F}_d$, including mixtures |
| Regularity used | Convexity/concavity, strict feasibility | Compactness, continuity, robust convexified separation |
| Comparison | Global scalar multiplier | Constant nonnegative matrix, global on $X$ |
| Safety restriction | Invariance of the given set | Prescribed degree and one fixed unsafe anchor |

The multiplier step in Lemma 2 is classical convex separation, closely related to the mechanism in (8); it is not claimed as new. The candidate contribution is the passage from robust convexified safety to a finite collection of feature-space facets, including a single unsafe separator. In general $\phi_d(f(x))$ does not induce a single-valued map on all of $Z$, so applying a deterministic invariance theorem to individual monomial trajectories would miss admissible mixtures.

**Broader context.** Robust converse barrier results already exist for discrete-time systems; see Liu [2], Section IV and Theorem 30. Geometric constructions of polyhedral invariants from contractive ellipsoids also precede this note [3]. Here the proposed characterization fixes the polynomial degree and concerns a convexified, potentially set-valued monomial relation with a fixed unsafe anchor. Whether this combination yields a substantively new theorem requires a fuller comparison of assumptions and conclusions with that literature.

## 4 Why the extra robustness is substantive

Let $X = [-1, 1]$, $f(x) = x$, $X_0 = [-1/2, 0]$, $X_u = [1/2, 1]$, and $d = 1$. The scalar polynomial

$$B(x) = x - \tfrac{1}{4}, \qquad A = [1]$$

satisfies (1), with both separation margins $1/4$. But for every $\varepsilon > 0$, $\mathcal{F}_1^\varepsilon$ permits $(s, s + h)$ whenever $s \in X$ and $0 \leq h \leq \varepsilon$. Starting at 0, finitely many such transitions reach $1/2$, staying in $X$. Therefore $\mathcal{K}_\varepsilon \cap J_1 \neq \varnothing$ for every $\varepsilon > 0$.

Thus an ordinary VBC can exist even when the robust hypothesis fails. Removing robustness from the converse requires a different argument; one must not infer the existence of a finite invariant polyhedron merely by approximating a possibly boundary-invariant convex set.

## 5 Adding functions at the same degree succeeds

Consider a contracting quarter-turn:

$$X = \{x \in \mathbb{R}^2 : \|x\|_2 \le 1\}, \quad f(x) = qRx, \quad q = \tfrac{9}{10}, \quad R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

$$X_0 = \{x : \|x\|_2 \le \tfrac{1}{4}\}, \qquad X_u = \{(\tfrac{3}{4}, 0)\}, \qquad d = 1.$$

The system maps $X$ into itself, and initial trajectories contract.

**One affine function is impossible.** Suppose $B(x) = a^\top x - c$ is a scalar VBC with multiplier $\lambda \ge 0$. Strict initial separation at 0 gives $c > 0$, while the unsafe anchor gives

$$c < \tfrac{3}{4}a_1 \le \tfrac{3}{4}\|a\|_2 < q\|a\|_2, \qquad a \ne 0.$$

At $x_* = R^\top a/\|a\|_2 \in X$, skew symmetry gives $a^\top x_* = 0$. Consequently

$$B(x_*) = -c < 0, \qquad B(f(x_*)) = q\|a\|_2 - c > 0,$$

contradicting $B(f(x_*)) \le \lambda B(x_*)$.

**Four affine functions succeed.** Take $r = 1/2$ and

$$B(x) = \begin{pmatrix} x_1 - r \\ x_2 - r \\ -x_1 - r \\ -x_2 - r \end{pmatrix}, \qquad A = q \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Direct substitution gives the global identity

$$B(f(x)) = AB(x) - \tfrac{1}{20}\mathbf{1}.$$

All components are at most $-1/4$ on $X_0$, and $B_1(3/4, 0) = 1/4$. The associated invariant is the square $[-1/2, 1/2]^2$. This proves a one-versus-four recovery at the same degree, without claiming that four is the minimum possible number.

The robust hypothesis also has a direct check. Since $\phi_1(x) = x$ and $f$ is linear, $\mathcal{F}_1$ is its graph over the unit disk. Set $C = [-r, r]^2$ and $\varepsilon = 1/100$. Then $I_1 + \varepsilon\mathbb{B}_V \subseteq C$. For a perturbed source $y = s + e \in C$ and successor $qRs + w$, with $\|e\|_2, \|w\|_2 \le \varepsilon$,

$$\|qRs + w\|_\infty \le qr + (q+1)\varepsilon = \tfrac{469}{1000} < \tfrac{1}{2}.$$

Thus $T_\varepsilon(C) \subseteq C$, and $C$ excludes $J_1$. Corollary 1 applies. This calculation includes every convexified transition pair and every subsequent convex combination.

## 6 An obstruction: adding same-degree functions cannot succeed

Reuse the nonlinear example from the earlier note:

$$f(x) = \tfrac{7}{8}x + \tfrac{5}{8}x^3 - \tfrac{1}{2}x^5, \qquad X = [-1, 1],$$
$$a = \tfrac{12}{25}, \quad u = \tfrac{1641}{3200}, \quad v = \tfrac{103}{200},$$
$$X_0 = [-a, a], \qquad X_u = [-v, -u] \cup [u, v].$$

Writing $z = x^2$ gives

$$f'(x) = \tfrac{1}{4} + (1 - z)(\tfrac{5}{8} + \tfrac{5}{2}z) \ge \tfrac{1}{4} \quad (x \in X);$$

together with $f(\pm 1) = \pm 1$, this verifies $f(X) \subseteq X$. There is no true unsafe trajectory from $X_0$: the single quadratic

$$b(x) = x^2 - \tfrac{1}{4}$$

is an *interpolation-inspired barrier certificate (IBC)* with one frame. If $b(x) \le 0$, then $f(x) = xh(x^2)$, where $h(z) = 1 + \frac{1}{2}(z - \frac{1}{4})(1 - z)$ lies in $[7/8, 1]$ for $0 \le z \le 1/4$. Hence $|f(x)| \le |x| \le 1/2$. The initial and unsafe margins are positive because $a < 1/2 < u$.

Nevertheless, at degree two put

$$\mu = \tfrac{369}{625}\delta_0 + \tfrac{128}{625}(\delta_{-3/4} + \delta_{3/4}), \qquad \mu_0 = \tfrac{1}{2}(\delta_{-a} + \delta_a), \qquad \nu = \tfrac{1}{2}(\delta_{-u} + \delta_u).$$

All are probability measures with support in their designated sets. Since $f(3/4) = 1641/2048$ and $u = af(3/4)/(3/4)$,

$$\int \phi_2 \,\mathrm{d}\mu = \int \phi_2 \,\mathrm{d}\mu_0 = (0, a^2), \qquad \int \phi_2 \circ f \,\mathrm{d}\mu = \int \phi_2 \,\mathrm{d}\nu = (0, u^2). \tag{9}$$

Thus $\big((0, a^2), (0, u^2)\big) \in \mathcal{F}_2$ has source in $I_2$ and successor in $J_2$. Already $C_0^1$ meets $J_2$, and so does $\mathcal{K}_\varepsilon$ for every $\varepsilon \geq 0$.

**Proposition 1** (No finite same-degree repair)**.** *For this example no forward fixed-anchor VBC of degree at most two exists, with any finite component count and any constant $A \geq 0$. The conclusion also holds through degree three.*

*Proof.* For any degree-at-most-two vector $B$, (9) and the unit masses imply

$$\int B \circ f \,\mathrm{d}\mu = \int B \,\mathrm{d}\nu \leq A \int B \,\mathrm{d}\mu = A \int B \,\mathrm{d}\mu_0 \leq 0.$$

The designated unsafe component has integral at least $\delta_u > 0$, a contradiction. All odd moments through degree three vanish by symmetry, so the same identities and argument apply to cubic components. □

The source atoms at $\pm 3/4$ lie outside $X_0$. Their mixtures imitate initial features while the successor mixtures imitate unsafe features. This is precisely the information that ordinary trajectory safety omits. The fixed-anchor restriction is essential: positivity of some component at each unsafe point does not in general imply positivity of one component averaged over $\nu$.

# 7 What this note establishes and what remains open

The note provides a proof of the candidate converse under its stated positive-margin condition, subject to your review. It constructs finitely many degree-bounded functions from robust safety of the convexified relation; the two analytical examples show that additional functions can repair a facet deficit but cannot repair a moment obstruction.

The unresolved step for the broader program is a converse for (1) without strict propagation. The identity example prevents replacing that question by the robust hypothesis without qualification. Also unresolved are a finite computational certificate for the closure assumption, quantitative facet bounds under additional geometry, and the full comparison with existing converse and invariant-approximation results. No new synthesis experiments are needed to evaluate the candidate proof.

**Code and reproducibility.** The computational materials for this note are available in the Certificate Expressiveness repository: `https://github.com/rezairaji60/certificate-expressiveness`. The repository includes `paper/converse.tex`, `src/exact_verify_converse.py`, and `evidence/converse_report.json`. The exact rational replay checks the affine comparison identity, the robustness bound, and the finite moment identities; it is not a machine verification of the general converse. The earlier degree-separation scripts and evidence are included in the same repository.

# References


[1] Z. Horváth, Y. Song, and T. Terlaky, *Invariance Conditions for Nonlinear Dynamical Systems*, arXiv:1607.01107v1, 2016. Theorem 3.10, equation (11), and Remark 3.11.

`https://arxiv.org/abs/1607.01107`.

[2] J. Liu, *Converse Barrier Functions via Lyapunov Functions*, IEEE Transactions on Automatic Control, vol. 67, no. 1, pp. 497–503, 2022. DOI: 10.1109/TAC.2021.3085419.

`https://arxiv.org/abs/2007.11086`.

[3] A. Alessio, M. Lazar, A. Bemporad, and W. P. M. H. Heemels, *Squaring the circle: An algorithm for generating polyhedral invariant sets from ellipsoidal ones*, Automatica, vol. 43, no. 12, pp. 2096–2103, 2007. DOI: 10.1016/j.automatica.2007.04.028.